

Entropy-Based Effective Degrees of Freedom in Principal Component Analysis

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May 27, 2026

Motivation

In many applications of principal component analysis (PCA), one encounters situations where:

- many eigenvalues are nonzero,
- but only a few contribute significantly,
- and one would like a quantitative measure of the *effective dimensionality*.

Instead of asking:

“What is the exact rank?”

we ask:

“How many degrees of freedom are effectively active?”

This leads naturally to entropy-based measures of dimensionality.

Principal Component Analysis

Consider a covariance matrix

$$C.$$

Diagonalizing the covariance matrix gives

$$C\mathbf{v}_i = \lambda_i\mathbf{v}_i,$$

where:

- λ_i are the eigenvalues,
- $\mathbf{v}_i$ are the principal components.

The eigenvalues represent the variance carried by each principal direction.

The total variance is

$$\sum_i \lambda_i.$$

Normalized Variance Fractions

Define normalized variance fractions:

$$p_i = \frac{\lambda_i}{\sum_j \lambda_j}$$

These quantities satisfy

$$\sum_i p_i = 1.$$

Interpretation:

- p_i behaves like a probability distribution,
- each component measures the fraction of total variance,
- dominant PCA modes correspond to large p_i .

Entropy of the Eigenvalue Spectrum

The Shannon entropy associated with the PCA spectrum is

$$S = - \sum_i p_i \log p_i$$

This entropy measures:

- how broadly distributed the variance is,
- how delocalized the spectrum is,
- how many modes contribute substantially.

Small entropy:

- variance concentrated in a few modes.

Large entropy:

- variance distributed among many modes.

Effective Number of Degrees of Freedom

The entropy itself is not directly an effective dimension.

Exponentiating the entropy gives:

$$N_{\text{eff}} = \exp \left(- \sum_i p_i \log p_i \right)$$

This quantity is interpreted as:

Interpretation

The effective number of principal components contributing significantly to the data.

This is also called:

- effective dimension,
- entropy dimension,
- effective rank

Limiting Case I: One Dominant Mode

Suppose

$$p_1 = 1, \quad p_{i>1} = 0.$$

Then

$$S = 0.$$

Hence

$$N_{\text{eff}} = e^0 = 1.$$

Interpretation:

- Only one effective degree of freedom exists.
- The data effectively lives in a one-dimensional manifold.

Limiting Case II: Uniform Distribution

Suppose k modes contribute equally:

$$p_i = \frac{1}{k}, \quad i = 1, \dots, k.$$

Then

$$S = - \sum_{i=1}^k \frac{1}{k} \log \frac{1}{k}.$$

This simplifies to

$$S = \log k.$$

Therefore

$$N_{\text{eff}} = e^{\log k} = k.$$

Thus the formula reproduces the intuitive number of active modes.

Why Exponentiate the Entropy?

Entropy itself is logarithmic.

Exponentiating converts the entropy into a quantity with units of:

“number of states”.

This is analogous to statistical mechanics:

$$S = \log \Omega,$$

where:

- S is entropy,
- Ω is the number of accessible microstates.

Thus

$$e^S$$

acts like an effective count of participating modes.

Connection with Statistical Mechanics

The PCA spectrum behaves similarly to occupation probabilities.

Analogy:

Statistical Mechanics	PCA
Microstates	Principal components
Occupation probabilities	Variance fractions p_i
Entropy	Spectral entropy
Accessible states	Effective dimension

This connection explains why entropy-based dimensionality measures are natural in physics.

Participation Ratio

Another commonly used measure is the participation ratio:

$$N_{\text{PR}} = \frac{1}{\sum_i p_i^2}$$

Comparison:

- entropy measure:

$$e^{-\sum_i p_i \log p_i},$$

- participation ratio:

$$\frac{1}{\sum_i p_i^2}.$$

Both quantify effective dimensionality.

The participation ratio penalizes dominant modes more strongly.

Applications in Physics

Entropy-based effective dimensions appear in many areas:

- random matrix theory,
- localization theory,
- condensed matter physics,
- quantum entanglement spectra,
- many-body physics,
- neuroscience,
- machine learning,
- signal processing.

In many-body physics:

- they measure the number of active collective modes,
- complexity of wave functions,
- degree of delocalization.

Effective Rank of a Matrix

In numerical linear algebra, the same quantity is often called the effective rank.

Given covariance matrix C :

$$\text{erank}(C) = \exp \left(- \sum_i p_i \log p_i \right)$$

Advantages compared to ordinary rank:

- robust against noise,
- insensitive to tiny eigenvalues,
- smoothly varying,
- physically meaningful.

Thus effective rank gives a continuous notion of dimensionality.

Interpretation of Different Spectra

Spectrum	Effective Dimension
One dominant eigenvalue	≈ 1
Two equal dominant modes	≈ 2
Broad spectrum	large
White noise spectrum	near full dimension

The entropy-based measure compresses the entire spectrum into a single scalar quantity.

Geometric Interpretation

The effective dimension characterizes:

- the intrinsic dimension of the data manifold,
- the number of active directions in feature space,
- the complexity of the covariance structure.

Low effective dimension:

- strong correlations,
- compressible data,
- collective behavior.

High effective dimension:

- weak correlations,
- distributed variance,
- noisy or high-complexity systems.

Summary

- PCA eigenvalues define variance fractions

$$p_i = \frac{\lambda_i}{\sum_j \lambda_j}.$$

- The Shannon entropy of the spectrum is

$$S = - \sum_i p_i \log p_i.$$

- Exponentiating the entropy gives the effective number of degrees of freedom:

$$N_{\text{eff}} = e^S.$$

- This quantity measures the effective dimensionality of the data.
- Closely related concepts:
 - participation ratio,
 - effective rank,
 - localization entropy,
 - entanglement entropy.

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